

ARCHITECTURE.md

System Architecture

High Level Flow

Developer ↓ Loadster CLI ↓ Scenario Parser ↓ Execution Engine ↓ Worker Pool ↓ HTTP Requests ↓ Metrics Collector ↓ Reporter

Component Architecture

CLI Layer

Responsibilities:

- Parse commands
- Parse flags
- Load config files
- Validate inputs

Suggested Libraries:

- Cobra
 - Viper
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Scenario Engine

Responsibilities:

- Read YAML
 - Parse stages
 - Build execution plan
 - Schedule load
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Worker Pool

Responsibilities:

- Manage Goroutines
- Maintain User Count
- Generate Requests
- Collect Responses

Architecture:

Controller ↓ Worker Manager ↓ Workers (N)

Each worker executes requests independently.

Metrics Engine

Responsibilities:

- Track Latencies
- Track Errors
- Track Success Rate
- Calculate Percentiles

Supported Metrics:

- Average
 - P50
 - P90
 - P95
 - P99
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Prometheus Layer

Expose:

localhost:9090/metrics

Metrics:

loadster_requests_total

loadster_errors_total

loadster_latency_seconds

loadster_active_users

loadster_rps

Distributed Mode

Master ↓ Kubernetes API ↓ Worker Pods

Master Responsibilities:

- Deploy Workers
- Assign Scenarios
- Collect Results

Worker Responsibilities:

- Generate Load
 - Send Metrics
 - Report Results
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Kubernetes Architecture

ArgoCD ↓ Helm Chart ↓ Loadster Namespace ↓ Master Deployment ↓ Worker Deployment ↓
ServiceMonitor ↓ Prometheus

CI/CD Architecture

GitHub Actions ↓ Lint ↓ Unit Tests ↓ Integration Tests ↓ Docker Build ↓ Trivy Scan ↓ Push GHCR ↓
ArgoCD Sync

Future Roadmap

v1

- Local CLI

v2

- Prometheus Integration

v3

- Distributed Kubernetes Mode

v4

- Grafana Dashboard

v5

- Multi Region Execution

v6

- SaaS Platform